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SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME:Artificial Intelligence

COURSE CODE: CSA1715

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3

Total Marks:30

Annexure A

Application Based Questions

Code of Conduct:

I _K.YAMINI(19257218)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Yamini

Annexure A

Application Based Questions

1. Intelligent Route Planning using A* and Greedy Search.

A smart delivery system is designed to transport medical supplies between cities. The system uses informed search strategies to determine optimal routes.

The road network is given below:

City	Connected To	Cost	Heuristic h(n) to Goal (G)
A	B, C	2, 4	7
B	D, E	3, 6	6
C	F	5	4
D	G	4	3
E	G	2	2
F	G	3	1
G	—	—	0

Tasks:

- Construct the search tree from A to G.
- Apply Greedy Best-First Search and show node expansion order.
- Apply A* algorithm using $f(n) = g(n) + h(n)$.
- Justify which algorithm is better for real-time emergency delivery systems.

2. Constraint Satisfaction Problem – University Timetable Scheduling

A university must schedule final exams for 5 subjects: AI, ML, DBMS, OS, CN across 3 days (Day 1, Day 2, Day 3) with the conditions, AI and ML cannot be scheduled on the same day, DBMS must be scheduled before OS, CN cannot be on Day 1, At most 2 exams per day, A student is enrolled in AI, DBMS, and CN → these cannot overlap

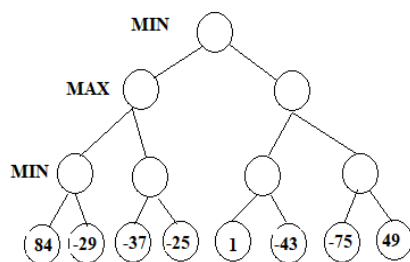
Tasks:

- Formulate this problem as a Constraint Satisfaction Problem (CSP) by defining, Variables, Domains and Constraints

- ii. Draw the constraint graph.
- iii. Solve using Backtracking Search.
- iv. Apply Forward Checking or Arc Consistency and explain how it reduces the search space.
- v. Provide one valid timetable solution.

3. Game Playing – Minimax and Alpha-Beta Pruning

An AI agent is designed to play a turn-based strategy game. The game tree is given below:



Tasks:

- i. Apply the Minimax algorithm and compute values for all internal nodes.
- ii. Identify the optimal move for MAX.
- iii. Apply Alpha-Beta Pruning to Show α and β values at each step and Clearly indicate which nodes are pruned
- iv. Compare nodes evaluated in Minimax vs Alpha-Beta
- v. Explain the importance of evaluation functions in complex games like chess.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Understanding of Problem Context	20% (2)	/2	/2	/2	/6
Application of Concepts	30% (3)	/3	/3	/3	/9
Problem-Solving Approach	20% (2)	/2	/2	/2	/6
Accuracy of Solution	20% (2)	/2	/2	/2	/6
Clarity	10% (1)	/1	/1	/1	/3
Total per Question	10 Marks	/10	/10	/10	/30

1. Intelligent Route Planning using A and Greedy Search*

1. Search Tree Construction

From given data:

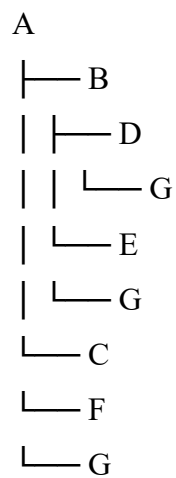
$A \rightarrow B (2), C (4)$

$B \rightarrow D (3), E (6)$

$C \rightarrow F (5)$

$D, E, F \rightarrow G$

Search Tree:



2. Greedy Best-First Search

Uses $h(n)$ only.

Heuristic values:

$A=7, B=6, C=4, D=3, E=2, F=1, G=0$

Node Expansion Order:

$A \rightarrow C \rightarrow F \rightarrow G$

Path Found:

$A \rightarrow C \rightarrow F \rightarrow G$

Cost:

$$4 + 5 + 3 = 12$$

3. A Algorithm $(f(n) = g(n) + h(n))^*$

Node	$g(n)$	$h(n)$	$f(n)$
A	0	7	7
B	2	6	8
C	4	4	8
D	5	3	8
E	8	2	10
F	9	1	10
G	9	0	9

Optimal Path:

$A \rightarrow B \rightarrow D \rightarrow G$

Total Cost:

$$2 + 3 + 4 = 9$$

4. Justification

- Greedy Search is faster but not optimal.
- A* ensures optimal path with minimum cost.
- In emergency delivery, correctness is more important than speed.

Best Algorithm: A*

- Guarantees shortest path
- Balances cost and heuristic
- Suitable for real-time critical systems

5. Conclusion

A* is the most efficient algorithm for intelligent route planning as it provides optimal and reliable solutions.

2. Constraint Satisfaction Problem – University Timetable

1. CSP Formulation

Variables:

AI, ML, DBMS, OS, CN

Domains:

{Day1, Day2, Day3}

Constraints:

- $AI \neq ML$
- DBMS before OS
- $CN \neq Day1$
- Max 2 exams per day
- AI, DBMS, CN cannot overlap

2. Constraint Graph

Edges:

- $AI \leftrightarrow ML$
- $DBMS \rightarrow OS$
- $AI \leftrightarrow DBMS$
- $DBMS \leftrightarrow CN$

3. Backtracking Solution

Try assignments:

Day1 → DBMS, ML

Day2 → AI, OS

Day3 → CN

All constraints satisfied

4. Forward Checking / Arc Consistency

- Removes invalid values early
- Reduces domain size
- Avoids unnecessary backtracking

Benefit:

Improves efficiency and reduces search space.

5. Valid Timetable

Day	Subjects
Day 1	DBMS, ML
Day 2	AI, OS
Day 3	CN

6. Conclusion

CSP with backtracking and forward checking efficiently solves scheduling problems.

3. Game Playing – Minimax and Alpha-Beta Pruning

1. Minimax Evaluation

Leaf nodes:

84, -29, -37, -25, 1, -43, -75, 49

Step 1 (MIN Level):

MIN1 $\rightarrow \min(84, -29) = -29$

MIN2 $\rightarrow \min(-37, -25) = -37$

MIN3 $\rightarrow \min(1, -43) = -43$

MIN4 $\rightarrow \min(-75, 49) = -75$

Step 2 (MAX Level):

MAX1 $\rightarrow \max(-29, -37) = -29$

MAX2 $\rightarrow \max(-43, -75) = -43$

Step 3 (Root MIN):

MIN $\rightarrow \min(-29, -43) = -43$

2. Optimal Move for MAX

MAX chooses branch leading to -29 (higher value)

3. Alpha-Beta Pruning

- α = best for MAX
- β = best for MIN

Pruning:

Some branches (like -75 subtree) are pruned when $\beta \leq \alpha$

Result:

Same result as minimax but faster

4. Comparison

Method	Nodes Evaluated	Efficiency
Minimax	All nodes	Slow
Alpha-Beta	Fewer nodes	Faster

5. Evaluation Function Importance

- Estimates board strength
- Helps in decision making
- Important for complex games like chess

6. Conclusion

Alpha-Beta pruning improves efficiency while maintaining optimal decision-making of minimax.